Solution overview

Book snapshot as-of 2026-10-31 · 10,000 customers · 14,685 accounts · observed default 16.1% · exposure $601.1m

How a risk analyst uses the dashboard

Portfolio

Morning book health: KPIs, default by product/region, employment × loan heatmap, predicted risk bands. Sidebar filters are the drill-down.

Account

Paste account_id. PD, risk band, observed default, bureau / repayment / collections drivers. Use this before calling a customer.

Anomalies

Work the 200-account queue. Read why it flagged. Confirm (collections) or dismiss (false positive). Open items remain listed.

Ask

Natural-language Q&A. Example: “Which loan type has the highest default rate?” Answer cites the grouping. Check the Portfolio chart before changing policy.

Mandatory AI features

1 Ask the Portfolio (LangGraph + Groq)

Trigger: Ask now on a natural-language question. False positive: a fluent rate for a cut that is not in facts.json — the validate node rejects it and we decline or fall back. False negative: an awkwardly phrased question the local retriever misses if Groq is down. Human step: read the cited grouping on the Portfolio tab before acting. No raw rows leave the app.

2 Anomaly detection

Trigger: IsolationForest tail and/or peer-extreme bureau DPD, write-off share, missed EMIs, collections intensity. False positive: a large mortgage with a one-off utilisation spike that is still paying on time. False negative: a two-month-old account with too little EMI history. Human step: confirm or dismiss on the review list; decision stored for QA.

Model results (stratified 75/25 on target_default)

CHAMPION LOGISTIC

AUC 0.667 KS 0.247 Gini 0.334

CHALLENGER LIGHTGBM

AUC 0.658 KS 0.246 Gini 0.317

SAMPLE

train 11,013 test 3,672 test DR 16.1%

LightGBM gain leaders (for explanation, not selection): loan_age_months, loan_amount, relationship_years, years_employed, interest_rate_pct, annual_income, repay_mean_dpd. Serve logistic as the production ranker until LightGBM beats it on Gini for two consecutive months. Monitor weekly PSI on income, loan amount, bureau DPD, write-off rate; monthly observed default vs PD band.

Top three portfolio insights

1. **Mortgage carries the highest default rate**
   Mortgage defaults at 24.6% vs Auto Loan at 13.9%. Price and limit unsecured products tighter; keep secured book as the ballast.
2. **Unemployed customers are the weakest employment segment**
   Default rate 35.0% versus Business Owner at 13.6%. Policy should require extra bureau quality or a lower LTI for this segment.
3. **Income band <25k has the highest default rate**
   39.6% default versus 120k+ at 7.9%. Affordability cuts belong in origination, not only collections.

Assumptions · what we need from bank IT and data to go live

Assumptions: snapshot as-of 2026-10-31; USD as provided; missing bureau DPD treated as 0 (not reported / current); missing interest rates imputed by product median; target_default is observed default, not a through-the-cycle PD; logistic is champion because it outranks LightGBM on this tape (Gini 0.334 vs 0.317). IT: scheduled warehouse extract, model registry, SSO in front of Streamlit, secrets vault for Groq, immutable audit of anomaly confirm/dismiss. Data: official default definition, vintage/closed-account policy, region hierarchy, monthly repayment tape, and a locked recent-origination hold-out month.